AERE 421 - Lab 1 - ANSYS Frame Analysis

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1. Introduction

We were tasked to use ANSYS to analyze a frame. First, the nodes were set, along with the beam support conditions. The nodes are the points that connect each beam. The support conditions are pinned for node 1 (at the origin) and a roller support for node 3 (bottom right). Once the nodes, beams, and support conditions were set, a –1500 N point load was applied to node 4 (the topmost point of the frame). Running the analysis in ANSYS provided calculations of the displacement at each node, the reaction forces at nodes 1 and 3, and the forces, stresses, and volume of each element. The values generated by ANSYS explain how the frame would react with a -1500 N point load applied in terms of reaction forces and displacement.

2. Data

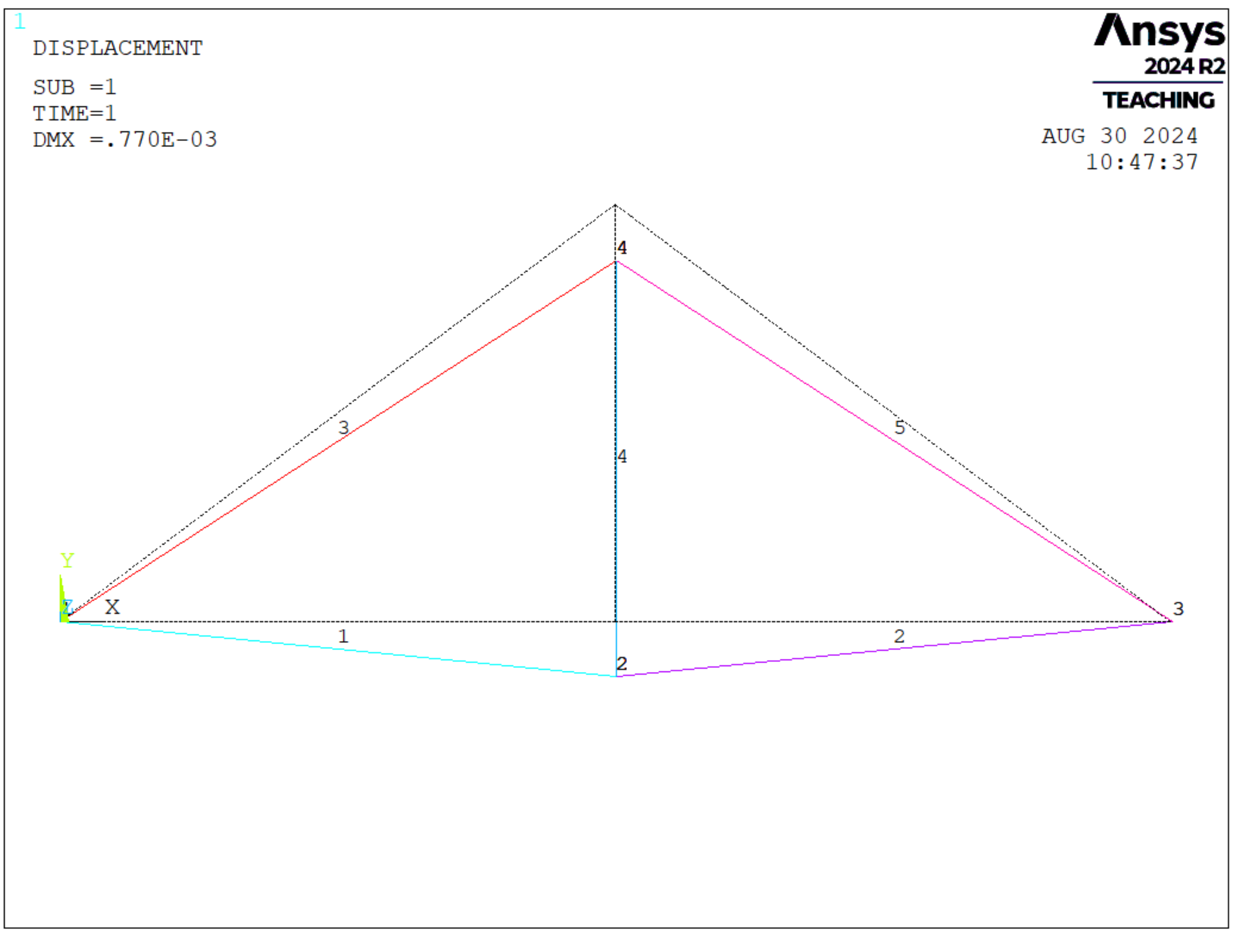

Figure 1. *Frame Displacement*

Node Number	Force X-Direction (N)	Force Y-Direction (N)	Force Z-Direction (N)
1	0.11369e-12	750.00	0.0000
2	0.0000	0.0000	0.0000
3	0.0000	750.00	0.0000
4	0.0000	0.0000	0.0000

Table 1. *Node Reaction Forces*

Node Number	Disp. X (m)	Disp. Y (m)	Disp. Z (m)	Disp. Vector Sum (m)
1	0.0000	0.0000	0.0000	0.0000
2	0.19048e-4	-0.76944e-3	0.0000	0.76968e-3
3	0.38095e-4	0.0000	0.0000	0.38095e-4
4	0.19048e-4	-0.76944e-3	0.0000	0.76968e-3

Table 2. *Node Displacement*

Element Number	Stress (kPa)	Axial Force (N)	Volume (m^3)
1	4000	1000	0.2500E-003
2	4000	1000	0.2500E-003
3	-25000	-1250	0.62500E-004
4	0	0	0.37500E-004
5	-25000	-1250	0.62500E-004

Table 3. *Element Stress, Force, and Volume*

3. Conclusion

From the data, the frame mainly behaved as anticipated. With a -1500 N point load acting on node 4, deflection of the frame downward is primarily expected. As seen in Table 3, the force distribution mostly aligns with what is logical. Member 4 (vertical center member) had no load and, in turn, no stress. Therefore, we conclude that member 4 is a zero force member, as it carries no load in the given configuration. Members 1 and 2 (bottommost horizontal members) took all their force in tension, which is intuitive due to the load applied and the support conditions. Both members are locked in the y-direction on the far ends but not in the middle. Member 4 is also splitting the two, as well as moving down. Therefore, the only option is tension in members 1 and 2.